

Order Diptera continued

Family Oestridae

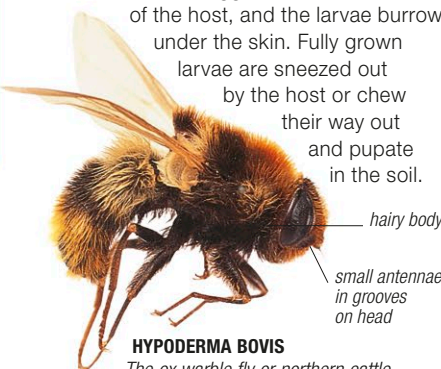
Bot flies and warble flies

Occurrence 70 spp. worldwide, especially in the Northern Hemisphere and Africa; near mammalian hosts



Many members of this heavy-bodied family look like bees. They are about $\frac{5}{16}$ –1 in (0.8–2.5 cm) long. Because the adults are short-lived and do not feed, the mouthparts are either tiny or absent. The larvae are all intestinal parasites of mammals, especially horses, goats, and camels. The females of some species lay larvae directly in the nostrils of hosts, where they feed and grow in the sinuses. In warble flies, eggs are laid on the hair

of the host, and the larvae burrow under the skin. Fully grown larvae are sneezed out by the host or chew their way out and pupate in the soil.



HYPODERMA BOVIS

The ox warble fly or northern cattle grub is common in the Northern Hemisphere, particularly near cattle. It damages hides, and reduces animal vigor and milk production.

Family Phoridae

Scuttle flies

Occurrence 3,000 spp. worldwide; in a wide variety of microhabitats

Habitat All

Also known as hump-backed flies, the brown, black, or yellowish scuttle flies have characteristically fast, jerky movements while running. They are $\frac{1}{64}$ – $\frac{1}{4}$ in (0.5–6 mm) long with a strongly down-turned head. The antennae appear to have only one segment and the hind femora are usually very stout. Conspicuous bristles are often present on the body; these look feathery when highly magnified. Adults live from a few days to a week, and may take fluids. Some of the larvae feed on fungi or decaying matter; others feed on insects, snails, and worms.



ANEVRINA THORACICA

Distributed over the Northern Hemisphere, the larvae of this fly favor soil, the corpses of animals, and moles' nests.

Family Psychodidae

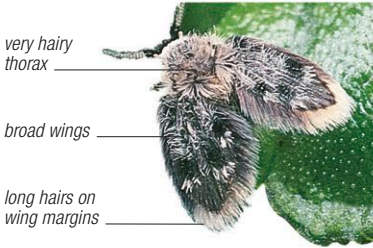
Sand flies and moth flies

Occurrence 1,000 spp. worldwide, especially in warmer regions; in water, rotting matter, and moist, shady habitats



These flies measure $\frac{3}{64}$ – $\frac{3}{16}$ in (1.5–5 mm) in length. Their bodies, wings, and legs are covered with long hairs or scales. The broad wings are held tentlike or partly spread in moth flies, or together above the body in sand flies. Sand and moth flies have a weak,

fluttery or hopping flight pattern. Female sand flies are blood-suckers and, particularly in the tropics, several species spread diseases.



PERICOMA FULIGINOSA

The larvae of this dark, widely distributed moth fly are, like sand flies, saprophagous. They develop in shallow water and in rot holes in trees.

Family Sarcophagidae

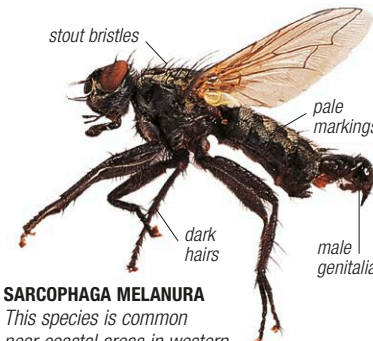
Flesh flies

Occurrence 2,300 spp. worldwide, especially in the Northern Hemisphere; foliage, flowers, carrion



Some of these robust-looking flies lay larvae in body cavities or wounds in vertebrates, including humans. They are mostly dull or silvery gray or black, and are $\frac{1}{16}$ – $\frac{3}{4}$ in (0.2–2 cm) long. The thorax typically has 3 dark longitudinal stripes on a gray background, and the abdomen appears checkered or spotted. Most females give birth to larvae that are laid or dropped in flight.

The larvae of many species feed on carrion, and some are parasitic on other insects, spiders, snails, and worms.



SARCOPHAGA MELANURA

This species is common near coastal areas in western Europe. Its larvae are found in rotting meat and may parasitize snails or insects.

Family Simuliidae

Black flies

Occurrence 1,600 spp. worldwide; larvae in flowing water, adults on plants near water



Black flies, also called buffalo gnats or turkey gnats, are $\frac{1}{32}$ – $\frac{3}{16}$ in (1–5 mm) long, and usually black, orange-red, or dark brown. Males drink nectar and water, while females feed on vertebrate blood. They lay their eggs on rocks or vegetation near water. The aquatic larvae, attached to submerged rocks or plants by a hooklike holdfast, filter food from the water with special bristles around the mouth.



SIMULIUM SP.

Although known as black flies, some members of this genus are not entirely black. Many species are carriers of the roundworm that causes river blindness in Africa and tropical America.

Family Syrphidae

Hover flies

Occurrence 6,000 spp. worldwide; common on flowers, especially umbelliferous ones



Hover flies include some of the most skilled fliers in the insect world. They feed on pollen and nectar, and can often be seen hovering over flowers, or darting after each other in a high-speed chase. Ranging from $\frac{3}{8}$ – $\frac{3}{4}$ in (1–2 cm) in length, they have large eyes and slender wings, and are often colored yellow or orange and black. This coloration—combined with their overall shape—makes them look very much like stinging bees or wasps, and deters many predators from attacking

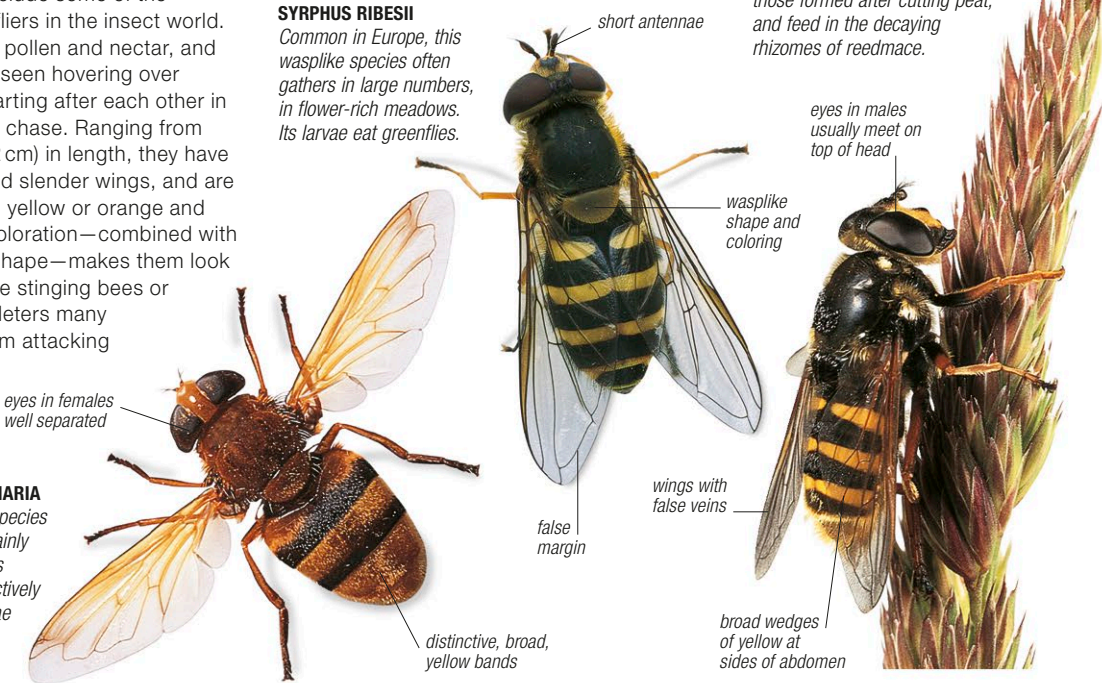
VOLUCELLA ZONARIA

This migratory species is distributed mainly over Europe. It is stout and distinctively banded. Its larvae scavenge inside wasps' nests.

them. Hover fly larvae are very varied, and live on a range of different foods. Some are predators, attacking aphids, sawfly larvae, and other soft-bodied insects, while others feed on rotting vegetation or animal dung. A few live as scavengers in the nests of bees and wasps, where they eat the dead larvae and pupae of their hosts.

SYRPHUS RIBESII

Common in Europe, this wasplike species often gathers in large numbers, in flower-rich meadows. Its larvae eat greenflies.



The larvae of one group, known as drone flies (*Eristalis* spp.), live in stagnant water, and use a telescopic, posterior "snorkel" to breathe.

SERICOMYIA SILENTIS

The common bog hoverfly is found on acid heathland in Europe. The larvae live in boggy pools, such as those formed after cutting peat, and feed in the decaying rhizomes of reedmace.